

Machine Learning Glossary (A-Z)

A

Accuracy

A statistic that calculates the percentage of all forecasts that are correctly classified.

Activation Function

To add non-linearity to neural networks, a mathematical function is applied to the output of a neuron.

B

Bias

Bias in machine learning can refer to either a parameter in neural networks that modifies the activation function or a mistake induced by model assumptions.

Backpropagation

An algorithm that modifies weights and propagates errors backward to train neural networks.

C

Classification

A task involving supervised learning in which the model forecasts a class or category label.

Clustering

A technique for unsupervised learning that clusters related data items.

D

Decision Tree

A prediction-making model that divides data into branches according to feature values.

Dimensionality Reduction

The procedure for lowering a dataset's input variable count while keeping crucial information intact.

E

Epoch

The training dataset is run through a machine learning model once.

Ensemble Learning

A method that enhances overall performance by combining several models.

F

Feature

A variable that a model uses as input.

Feature Engineering

Process of adding or changing elements to enhance the performance of the model.

G

Gradient Descent

An optimization approach that reduces error by updating parameters.

Generalization

The capacity of a model to function well with unknown data.

H

Hyperparameters

Pre-training settings that govern the behavior of the model.

Hidden Layer

Input-processing middle layer in a neural network.

I

Input Layer

The neural network's initial layer, where information is introduced.

Imbalanced Dataset

A dataset in which certain classifications are significantly more prevalent than others.

J

Jaccard Similarity

A way to compare how similar two sets are to one another.

Jupyter Notebook

A popular interactive coding environment for machine learning projects.

K

K-Nearest Neighbors (KNN)

algorithm that uses the closest neighbors to categorize data.

K-Means Clustering

Data is grouped into K clusters using an unsupervised technique.

L

Loss Function

A function that calculates the inaccuracy of model predictions.

Logistic Regression

A classification algorithm that uses the logistic function to forecast probabilities.

M

Model

A prediction-trained mathematical representation.

Mean Squared Error (MSE)

A loss function that calculates the average squared discrepancy between the actual and anticipated values.

N**Neural Network**

A group of interrelated layers that were modeled after the structure of the human brain.

Normalization

Adjusting input characteristics to a standard range.

O**Overfitting**

When a model performs poorly on new data but learns training data too well.

Outlier

A piece of data that differs greatly from the others.

P**Precision**

The percentage of actual favorable results among all anticipated positive results.

Principal Component Analysis (PCA)

A dimensionality reduction technique that finds new feature axes.

Q**Quantization**

Use discrete values to approximate continuous values in order to reduce the size of the model.

Query

A query to a database or dataset for information.

R

Recall

The percentage of detected true positives among all actual positives.

Reinforcement Learning

A teaching strategy in which agents pick up behaviors through rewards and penalties.

S

Supervised Learning

Using labeled input-output pairs to train a model.

Support Vector Machine (SVM)

A classifier that uses the optimal hyperplane to divide classes.

T

Training Data

Data used to train a machine learning model.

Transfer Learning

Applying a previously trained model to a novel but connected problem.

U

Unsupervised Learning

Unlabeled data can be used to uncover hidden patterns.

Underfitting

When a model is too basic to identify patterns in the data.

V

Validation Set

A dataset for adjusting the hyperparameters of the model as it is being trained.

Variance

Error brought on by the model's sensitivity to minor training adjustments.

W**Weights**

Neural network parameters that change input data.

Word Embedding

Words are represented in NLP as dense vectors.

X**XGBoost**

A gradient boosting library that is optimized for high-performance machine learning.

XML Data

For organized datasets, a markup language format is occasionally utilized.

Y**Y-Label (Target Variable)**

A supervised machine learning model's predicted output or label.

Yield Curve Modeling (ML in Finance)

ML is used to forecast changes in interest rates.

Z**Zero-Shot Learning**

Using past information, the model forecasts classes it has never seen before.

Z-Score Normalization

By deducting the mean and dividing by the standard deviation, data can be standardized.